Business Calculus (MATH 2350)
Summer 2018
Thursday, 06/07/2018
Quiz 4: Section 11.4,11.5

Name (Print): ____________________

Score: / 100

Section 11.4, 11.5

1. $\boxed{\text{20 points}}$ Let y represent a function of x. Derive a general formula for $\frac{d}{dx}(x^n y^m)$. Show all of your steps.

$$\begin{aligned} \frac{d}{dx}(x^n y^m) &= \frac{d}{dx}(x^n)y^m + x^n\frac{d}{dx}(y^m) \quad \text{Product Rule} \\ &= (nx^{n-1})y^m + x^n(my^{m-1}\frac{dy}{dx}) \quad \text{Chain Rule (since y is a function of x)} \\ &= nx^{n-1}y^m + mx^n y^{m-1}\frac{dy}{dx} \quad \text{Simplify} \end{aligned}$$

2. $\boxed{\text{10 points}}$ Find $\frac{d}{dy}\left(e^{2y^2+y+1}\right)$.

Hint: Make sure you read the question carefully.

Since we are differentiating with respect to y, we need not worry about any of the concepts from implicit differentiation. Thus, we yield:

$$\begin{aligned} \frac{d}{dy}\left(e^{2y^2+y+1}\right) &= e^{2y^2+y+1}\frac{d}{dy}(2y^2+y+1) \quad \text{Chain rule} \\ &= e^{2y^2+y+1}(4y+1) \end{aligned}$$

3. The total cost (in hundreds of dollars) of producing x auto body frames is

$$C(x) = 5 + (2x+3)^{1/3} + 2x.$$

(a) $\boxed{\text{10 points}}$ Find $C'(x)$.

Using the chain rule, we yield:

$$\begin{aligned} C'(x) &= \left(5 + (2x+3)^{1/3} + 2x\right)' \\ &= \left((2x+3)^{1/3}\right)' + 2 \\ &= \frac{1}{3}(2x+3)^{-2/3}(2x+3)' + 2 \quad \text{Chain rule} \\ &= \frac{2}{3}(2x+3)^{-2/3} + 2 \end{aligned}$$

(b) $\boxed{\text{10 points}}$ Find $C'(12)$.

Evaluate your result from the previous section:

$$\begin{aligned} c'(12) &= \frac{2}{3}(2(12)+3)^{-2/3} + 2 \\ &= \frac{2}{3}(27)^{-2/3} + 2 \\ &= \frac{2}{3}(3^3)^{-2/3} + 2 \\ &= \frac{2}{3\sqrt[3]{(3^3)^2}} + 2 \\ &= \frac{2}{3\sqrt[3]{3^6}} + 2 \\ &= \frac{2}{3(3^2)} + 2 \\ &= \frac{2}{3^3} + 2 \\ &= \frac{2}{27} + \frac{2(27)}{27} \\ &= \frac{2+2(27)}{27} = \frac{56}{27} \end{aligned}$$

4. For the following problems, find find $\dfrac{dy}{dx}$.

   (a) $\boxed{\text{10 points}}$ $x = 4y + 12$

$$
\begin{aligned}
\frac{d}{dx}(x) &= \frac{d}{dx}(4y+12)\\
1 &= 4\frac{d}{dx}(y) + \frac{d}{dx}(12)\\
1 &= 4\frac{dy}{dx}\\
\frac{dy}{dx} &= \frac{1}{4}
\end{aligned}
$$

   (b) $\boxed{\text{10 points}}$ $\dfrac{1+x^2}{1+y^2} = 1$

$$
\begin{aligned}
\frac{d}{dx}\left(\frac{1+x^2}{1+y^2}\right) &= \frac{d}{dx}(1)\\
\frac{\frac{d}{dx}(1+x^2)(1+y^2) - (1+x^2)\frac{d}{dx}(1+y^2)}{(1+y^2)^2} &= 0 \quad \text{Quotient rule}\\
\frac{(2x)(1+y^2) - (1+x^2)(2y\frac{dy}{dx})}{(1+y^2)^2} &= 0 \quad \text{Chain Rule}\\
(2x)(1+y^2) - (1+x^2)(2y\frac{dy}{dx}) &= ((1+y^2)^2)(0)\\
2x(1+y^2) - (1+x^2)(2y\frac{dy}{dx}) &= 0\\
-(1+x^2)(2y\frac{dy}{dx}) &= -2x(1+y^2)\\
(2y\frac{dy}{dx}) &= \frac{-2x(1+y^2)}{-(1+x^2)}\\
\frac{dy}{dx} &= \frac{-2x(1+y^2)}{-2y(1+x^2)}\\
\frac{dy}{dx} &= \frac{x(1+y^2)}{y(1+x^2)} \quad \text{This is an acceptable final answer}\\
\frac{dy}{dx} &= \frac{x}{y} \quad \text{Via the identity given in the question}
\end{aligned}
$$

(c) $\boxed{\text{10 points}}$ $ln(x^5) = e^y$

$$
\begin{aligned}
\frac{d}{dx}(ln(x^5)) &= \frac{d}{dx}(e^y) \\
\frac{d}{dx}(5ln(x)) &= \frac{d}{dx}(e^y) \\
5\frac{d}{dx}(ln(x)) &= e^y\frac{dy}{dx} \quad \text{Chain Rule} \\
\frac{5}{x} &= e^y\frac{dy}{dx} \\
\frac{dy}{dx} &= \frac{5}{xe^y}
\end{aligned}
$$

(d) $\boxed{\text{10 points}}$ $x^2e^y = y + x^2$

$$
\begin{aligned}
\frac{d}{dx}(x^2e^y) &= \frac{d}{dx}(y + x^2) \\
\frac{d}{dx}(x^2)(e^y) + (x^2)\frac{d}{dx}(e^y) &= \frac{d}{dx}(y) + \frac{d}{dx}(x^2) \quad \text{Product Rule} \\
(2x)(e^y) + (x^2)(e^y\frac{dy}{dx}) &= \frac{dy}{dx} + 2x \quad \text{Chain Rule} \\
(x^2)(e^y\frac{dy}{dx}) - \frac{dy}{dx} &= 2x - (2x)(e^y) \\
(x^2e^y - 1)\frac{dy}{dx} &= 2x(1 - e^y) \\
\frac{dy}{dx} &= \frac{2x(1 - e^y)}{(x^2e^y - 1)}
\end{aligned}
$$

(e) $\boxed{\text{10 points}}$ $x^2 + \ln(1/y) = 24$

$$\frac{d}{dx}(x^2 + \ln(1/y)) = \frac{d}{dx}(24)$$
$$\frac{d}{dx}(x^2) + \frac{d}{dx}(\ln(1/y)) = 0$$
$$2x + \frac{1}{1/y}\frac{d}{dx}(1/y) = 0 \quad \text{Chain Rule}$$
$$2x + y\frac{-\frac{dy}{dx}}{y^2} = 0$$
$$2x + \frac{-\frac{dy}{dx}}{y} = 0$$
$$\frac{-1}{y}\frac{dy}{dx} = -2x$$
$$\frac{dy}{dx} = 2xy$$

5. $\boxed{\text{15 points (bonus)}}$ If $f(x) = \dfrac{x+1}{(x+3)^2}$, find the value(s) of x where the tangent line of f is horizontal.

First, we calculate $f'(x)$:

$$\begin{aligned}
f'(x) &= \left(\frac{x+1}{(x+3)^2}\right)' \\
&= \frac{(x+1)'(x+3)^2 - (x+1)((x+3)^2)'}{((x+3)^2)^2} \quad \text{Quotient Rule} \\
&= \frac{(x+3)^2 - (x+1)(2(x+3)(x+3)'}{(x+3)^4} \quad \text{Chain Rule} \\
&= \frac{(x+3)^2 - (x+1)(2(x+3)}{(x+3)^4} \\
&= \frac{(x+3)((x+3) - 2(x+1))}{(x+3)^4} \\
&= \frac{(x+3) - 2(x+1)}{(x+3)^3} \\
&= \frac{-x+1}{(x+3)^3}
\end{aligned}$$

Now, solve $f'(x) = 0$.

$$\begin{aligned}
f'(x) &= 0 \\
\frac{-x+1}{(x+3)^3} &= 0 \\
-x+1 &= 0(x+3)^3 \\
-x+1 &= 0 \\
x &= 1
\end{aligned}$$

Thus, as the derivative at a point is the slope of the tangent line at that point, the slope is flat (i.e. has derivative zero) when $x = 1$.